

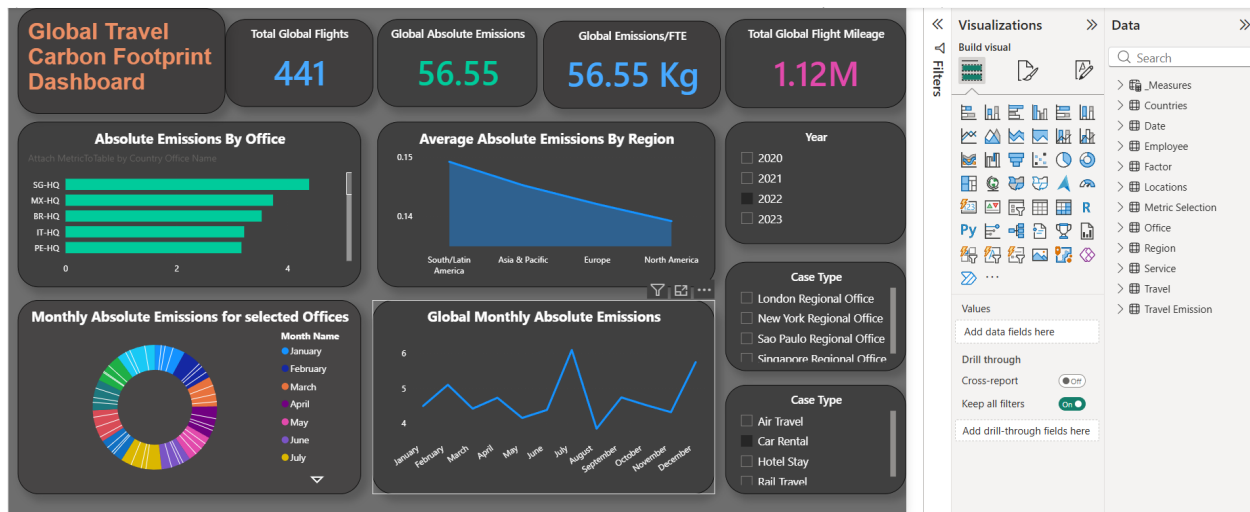
GLOBAL TRAVEL CARBON FOOTPRINT & EMISSIONS INSIGHTS DASHBOARD

Business Travel GHG (Scope 3) Analytics in Power BI

Project Documentation

Prepared as a Data Analytics Portfolio Project

Tool: Microsoft Power BI Desktop



1. Project Overview

This project simulates a multinational organization's business-travel carbon footprint program and presents an interactive Power BI dashboard that tracks flight-related greenhouse gas (GHG) emissions across offices, regions, and travel purpose. It demonstrates end-to-end BI development skills: relational data modeling across 11 tables, DAX time-intelligence and emissions calculations, interactive filtering, and drill-down navigation for sustainability (ESG) reporting.

The dashboard answers questions a sustainability or travel operations manager would actually ask: How much CO₂e are we emitting from business travel, and is it going up or down? Which regions and offices contribute the most? How much of our footprint comes from billable client work versus internal administration? Are our short, medium, and long-haul flights contributing proportionally to our emissions?

2. Business Objective

The dashboard was built to help an ESG / travel operations team:

- Monitor total absolute emissions (kg CO₂e) and emissions per employee (FTE) in real time.
- Compare emissions across regions (North America, Europe, Asia & Pacific, South/Latin America) and drill into individual offices.
- Break down emissions by haul type (Short, Medium, Long) to understand which flight categories drive the footprint.
- Segment emissions by case/engagement type (Billable, Administrative, Client Development, Pro Bono) to separate client-driven travel from internal travel.
- Track month-over-month and year-over-year emissions trends to support carbon-reduction target setting.

- Support inter-office and own-office emissions comparisons for internal sustainability benchmarking.

3. Dataset Description

The dataset is a synthetic global business-travel and emissions log spanning 11 related tables (GHG_Emissions_Dashboard_Dataset.xlsx), modeled on a real-world corporate travel and ESG data architecture:

Table	Description
Travel	One row per flight leg: carrier, haul class, mileage, cost, dates, links to employee/office/service
Travel Emission	One row per flight leg's calculated emissions: emission factor used, absolute emission (kg CO2e), case type
Employee	Employee master: department, status, home country
Office	Office/legal-entity master: billing office, GL region/country, cost center
Region	4 global regions with regional ESG coordinator and preferred currency
Countries	20 countries with ESG coordinator and country office contact
Locations	City-level office locations with coordinates
Service	Travel service type: Air, Rail, Car Rental, Hotel
Factor	Emission factors (kg CO2e per passenger-km) by haul class and year
Date	Standalone calendar table (2020-2023) for time intelligence
Metric Selection	Disconnected what-if table driving the Absolute Emission / Absolute Emission-per-FTE toggle

Dataset size: 1,500 travel/emission records spanning four full annual cycles (2020-2023), across 20 countries, 20 offices, and 120 employees.

4. Tools & Skills Demonstrated

Area	Details
BI Tool	Microsoft Power BI Desktop
Data Modeling	11-table relational model (star/snowflake hybrid) with a dedicated Date table for time intelligence
DAX	Custom measures for absolute emissions, emissions/FTE, year-over-year comparisons, rolling 3-year windows, and a field-parameter-style metric toggle (see Section 6)
Visuals Used	KPI cards, area chart, stacked column chart, clustered bar chart, donut chart, line/column combo charts
Interactivity	Multi-select slicers (Haul, Comparison Year, Travel Start Date, Case Type, Operating Office, Year, Region), cross-report drill-through
Navigation	Multi-page report: Inter-Office Emissions and Own-Office Emissions views
Design	Dark theme with gold/teal/blue accent palette, card-based KPI layout, geographic and time-based comparisons

5. Data Model

The model follows a snowflaked star schema:

- **Fact table:** Travel Emission — one row per flight leg's emission calculation, holding Emission Factor, Absolute Emission, Case Type, and Date ID.
- **Bridge/fact table:** Travel — one row per flight leg, holding carrier, haul class, mileage, cost, and links to Employee, Office, Service, Region, and Country.
- **Dimension tables:** Date, Countries, Region, Office, Locations, Employee, Service, Factor, and the disconnected Metric Selection table.
- **Key relationships:** Travel Emission[Travel Code] to Travel[Travel Code] (one-to-one); Travel[Office

Code] to Office[Office Code]; Office[Country Code] to Countries[Country Code]; Countries[Region Code] to Region[Region Code]; Travel Emission[Date ID] to Date[Date ID]; Travel Emission[Factor Code] to Factor[Factor Code].

Keeping emissions in a separate fact table from the raw travel/cost data (rather than one flat table) keeps the emissions-factor lookup clean per haul class and year, and mirrors how real ESG reporting systems separate activity data from the emissions calculation applied to it.

6. Key DAX Measures

Measure	Purpose
Total Global Flights	COUNTROWS of the Travel table — overall travel volume
Global Absolute Emissions	SUM of Travel Emission[Absolute Emission] — total kg CO2e
Global Emissions/FTE	Global Absolute Emissions ÷ DISTINCTCOUNT of active employees
Total Global Flight Mileage	SUM of Travel[Flight Mileage] across all legs
SumEmissionFactor	SUM of Travel Emission[Emission Factor] for the current filter context
_This Year / _Previous Year	CALCULATE of Absolute Emission sliced to the current vs. prior calendar year, for year-over-year comparison cards
_Last 3 Years / _Next 3 Years	Rolling 3-year window measures using DATESINPERIOD, for trend framing
_Year to date	CALCULATE(Absolute Emission, DATEADD(Travel[Travel Start Date], -12, DAY)) — trailing 12-month emissions
Travel Emission/FTE	Emissions measure normalized per employee, toggled via the Metric Selection what-if parameter

7. Dashboard Pages

Page 1 — Inter-Office Emissions (Landing Page)

The main analysis page gives a full global operational snapshot:

- KPI strip: Total Global Flights, Global Absolute Emissions, Global Emissions/FTE, Total Global Flight Mileage — an at-a-glance sustainability health check.
- Absolute Emissions by Office (bar): identifies the highest-emitting offices for targeted intervention.
- Average Absolute Emissions by Region (area): North America, Asia & Pacific, Europe, South/Latin America compared side by side.
- Monthly Absolute Emissions for Selected Offices (donut): month-by-month share for a filtered set of offices.
- Global Monthly Absolute Emissions (column): trend across the year, useful for spotting seasonal travel spikes.
- Slicer panel: filter the entire page by Operating Office, Case Type, and Year.

Page 2 — Own-Office Emissions

A dedicated page for single-office / haul-level analysis:

- Monthly Absolute Emissions (column): trend for the currently filtered office or scope.
- Case Type breakdown (bar): Billable vs. Pro Bono vs. Administrative vs. Client Development share of emissions.
- Haul-class stacked column: Long / Medium / Short Haul emissions by month, with a Comparison Year slicer for year-over-year framing.
- Slicer panel: filter by Haul, Comparison Year, Travel Start Date (Last/Next 3 Years, This/Previous Year), and Case Type.

8. Key Insights

Based on the current dataset (1,500 travel/emission records, 2020-2023):

- **Overall footprint:** 630,365 kg CO₂e (≈630 tonnes) of total absolute emissions across 4,491,494 km of flown distance, averaging roughly 5,253 kg CO₂e per employee (FTE).
- **Regional split:** Europe (196,399 kg) and Asia & Pacific (185,270 kg) are the two largest contributors, followed by South/Latin America (146,366 kg) and North America (102,330 kg) — a useful starting point for setting region-specific reduction targets.
- **Haul-class impact:** Long-haul flights account for the largest emissions share (288,897 kg from just 304 legs) despite being the least frequent flight type — each long-haul trip carries roughly 3-4x the footprint of a short-haul trip, making long-haul reduction the highest-leverage lever.
- **Case-type mix:** Billable client travel drives the majority of the footprint (346,921 kg, 821 legs), with Client Development (127,151 kg) a distant second — Administrative and Pro Bono travel together are a relatively small share, suggesting most of the footprint is tied directly to revenue-generating work.
- **Office concentration:** The Mexico, Peru, Germany, United States, and Italy offices are the top five emitters, each in the 34,000-37,000 kg range — a fairly even spread rather than one dominant outlier, meaning reduction efforts likely need to be broad-based rather than targeted at a single site.
- **Year-over-year trend:** Annual emissions have stayed relatively flat across the four years (~150,000-167,000 kg per year, peaking in 2022), suggesting travel volume recovery after 2020-2021 rather than a clear reduction trend — a natural narrative hook for a "where do we go from here" sustainability discussion.

9. How to Use the Dashboard

- Use the slicers (Operating Office, Case Type, Year, Haul, Comparison Year) to filter every visual on a page simultaneously.
- Hover over any chart for tooltip-level detail (exact kg CO₂e, mileage, and flight counts).
- Use the Metric Selection toggle to switch KPI cards between Absolute Emission and Absolute Emission/FTE views.
- Switch between the Inter-Office Emissions and Own-Office Emissions pages using the page tabs to move from a global view to a single-office deep dive.

10. Potential Enhancements

- Add a Reduction Target line/band overlay comparing actual emissions against a stated annual carbon target.
- Add a Top N Highest-Emitting Routes table (origin-destination pairs) for targeted travel-policy interventions.
- Introduce ground-transport and hotel-stay emissions (currently Air-only) to build a fuller Scope 3 travel picture.
- Publish to the Power BI Service with a scheduled refresh connected to a live travel-booking or expense system (e.g., Concur, SAP Concur, Egencia).
- Add row-level security (RLS) so each regional ESG coordinator only sees their own region's data.

11. Project Files

File	Description
GHG_Emissions_Dashboard_Dataset.xlsx	Source data (11 related tables, 1,500 travel/emission records) used to build the model

File	Description
Global_Travel_Emissions_Dashboard.pbix	Power BI report file containing the data model, DAX measures, and both report pages
Project Documentation (this file)	Overview, methodology, and insights for portfolio presentation

12. Suggested Dashboard Names

A few naming options considered for this project, ranked by how well they communicate business value to a portfolio reviewer:

- **Global Travel Carbon Footprint Dashboard** — recommended. Leads with the business metric (carbon footprint) hiring managers and sustainability leads scan for.
- **Business Travel Emissions Insights** — clean, functional, and easy to search/find; pairs well with an ESG/sustainability job description.
- **CarbonTrack: Corporate Travel Emissions Dashboard** — more brand-like, good if presenting as a product concept.
- **GHG Emissions Analytics: Travel Footprint & SLA-Style Reduction Insights** — descriptive, strong for a portfolio website or resume bullet.
- **Sustainability Ops Command Center** — enterprise, executive-facing tone, mirrors ESG steering-committee reporting.